Bedrock

Real-time ICU early warning on IBM LinuxONE. It catches more deteriorating patients with fewer alarms, and patient data never leaves the machine.

- Theme: Real-time AI for critical decisions
- with AI secured: innovation without exposure

THE PROBLEM

ICU nurses face too many alarms, and the alarms still miss patients who are getting worse.

Wards score deterioration with NEWS2, a points table from vital signs. It looks at one moment, it fires constantly, and it misses a large share of patients who go on to deteriorate. Staff learn to tune alarms out.

1.81

NEWS2 ≥ 5 alerts **per bed per day**

30%

of patients who develop sepsis get **no warning** from NEWS2 before it starts

6,053

held-out ICU stays we measured this on, **449** of them developed sepsis

Data: PhysioNet/CinC Challenge 2019 (hospitals A, B), PhysioNet 2012 (M), eICU Collaborative Research demo (C, external validation). De-identified research data.

THE SOLUTION

Five risks per bed, every hour, ranked so the sickest patient is always at the top.

Sepsis
next 6 h

Hypotension
next 6 h

Acute kidney injury
next 24 h

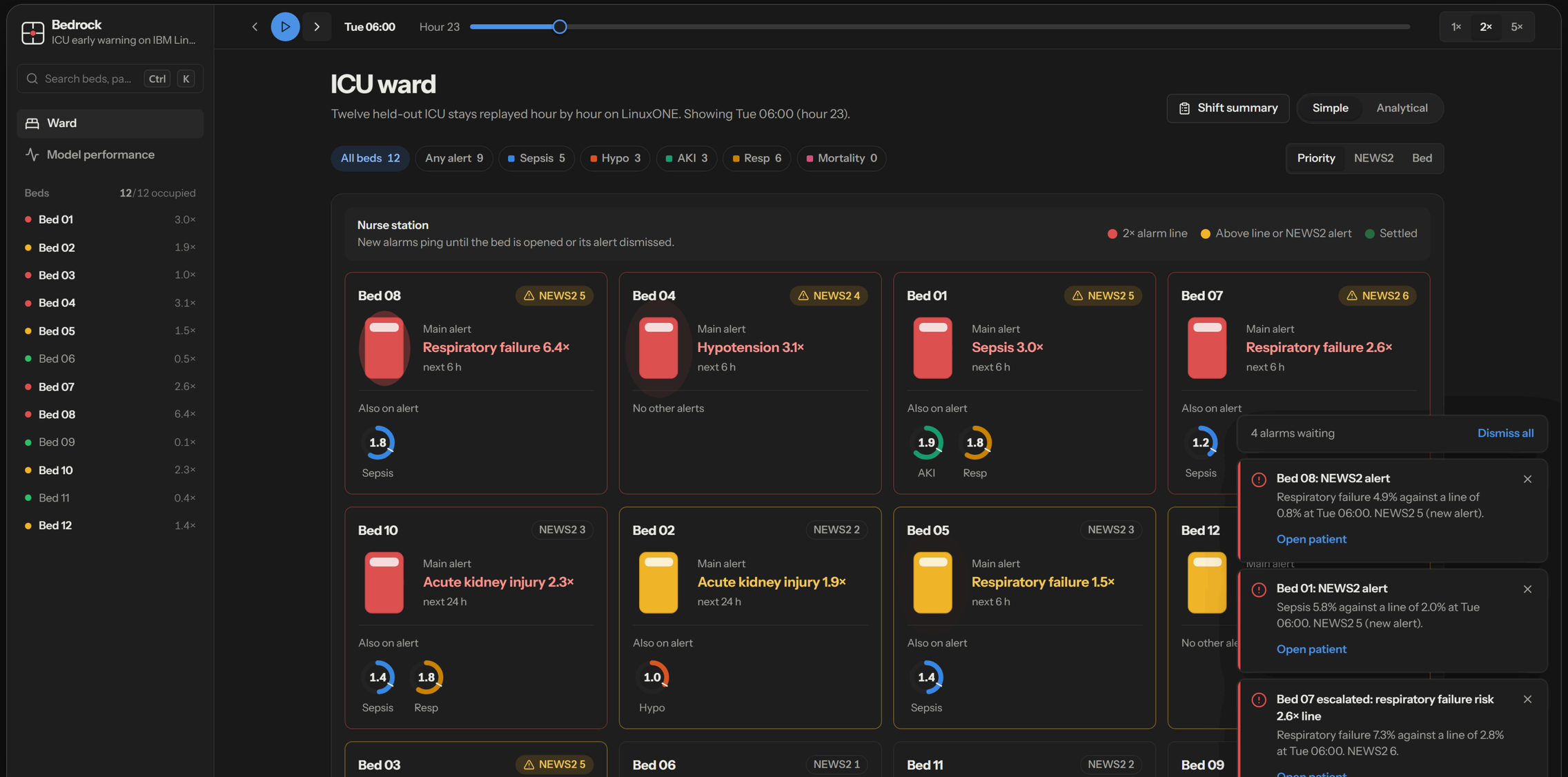
Respiratory failure
next 6 h

Mortality
this stay

- **Five XGBoost heads** served with Snap ML on LinuxONE, sharing one 107-feature pass per bed.
- **Fair alarm lines:** each head's threshold raises the same number of alarms as NEWS2 ≥ 5, so every comparison is like for like.
- **Explained:** every alarm shows its main drivers, and Granite 4.0 writes a checked SBAR handover.
- **Built for the ward:** nurse station alarms with acknowledgement and escalation, a simple and an analytical view, and a shift summary.

DEMO · LIVE ON LINUXONE

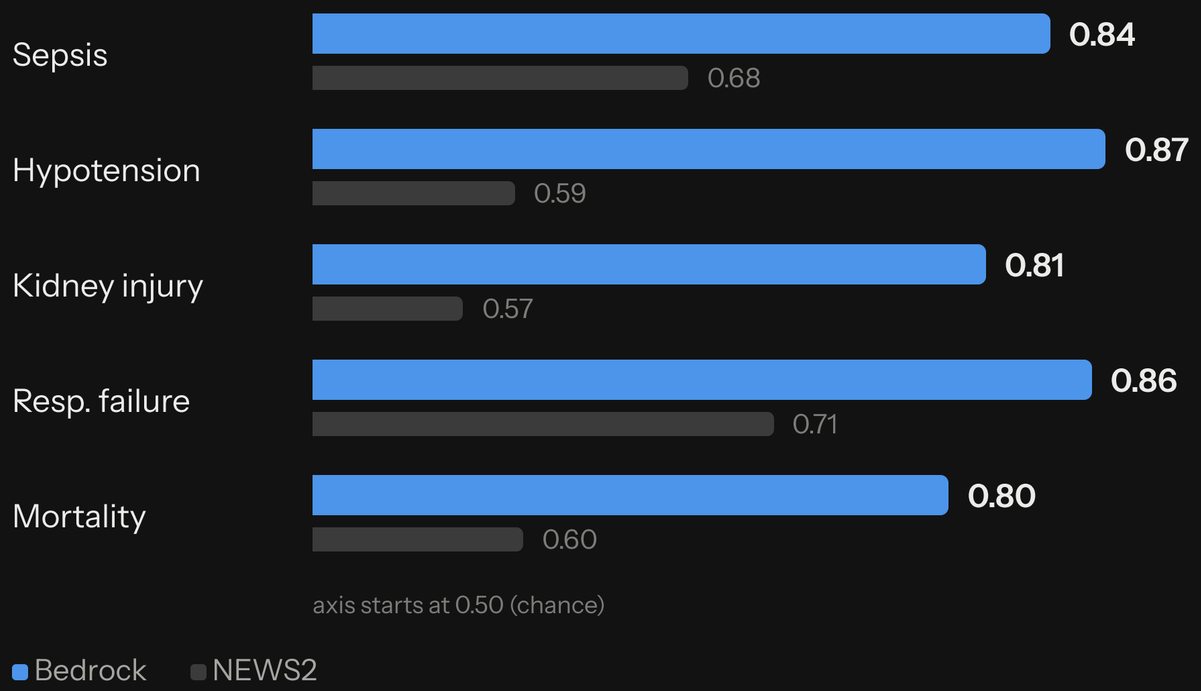
The ward board: real held-out ICU stays replayed hour by hour.



RESULTS ON HELD-OUT PATIENTS

Same catch rate as NEWS2 with 78% fewer alarms.

AUROC: Bedrock vs NEWS2



SAME CATCH RATE (70%)

0.39 vs 1.81

alerts per bed per day: **78% fewer alarms**

SAME ALARM BUDGET (1.81 PER BED-DAY)

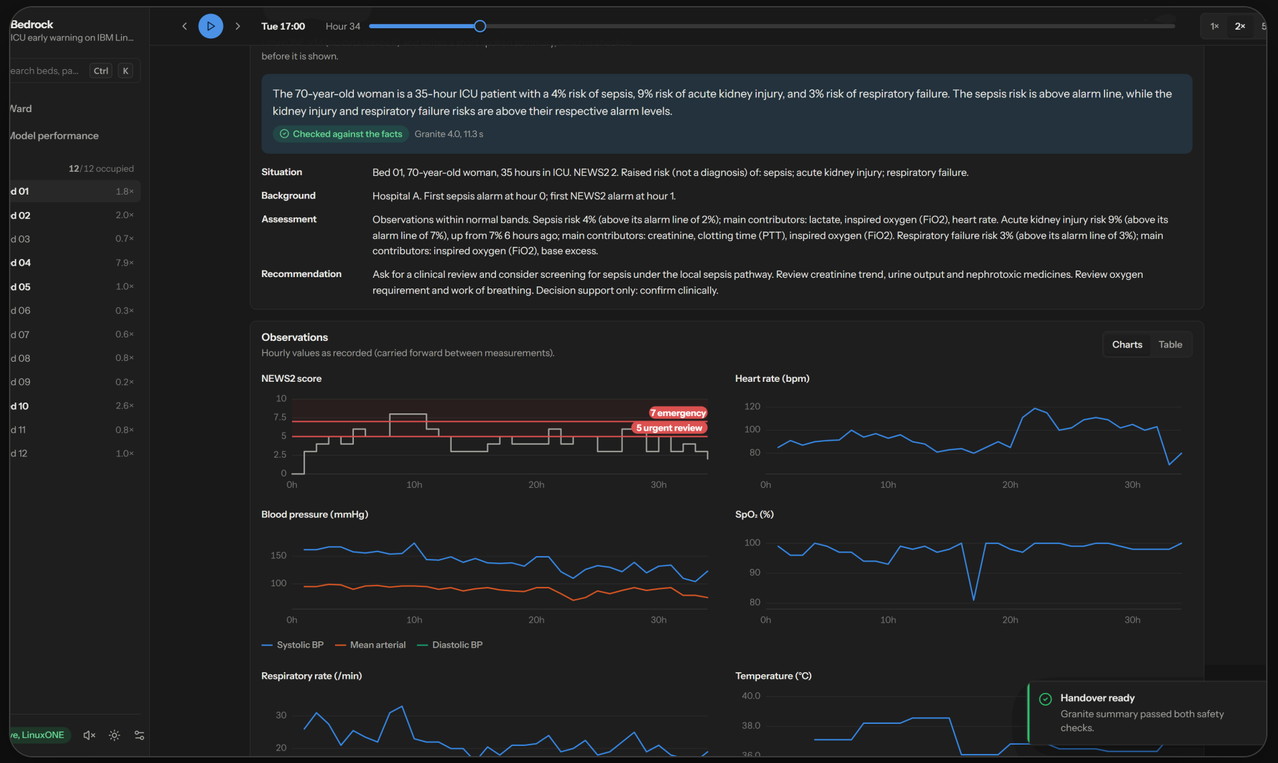
95% vs 70%

of patients who develop sepsis are warned before onset

Sepsis head, 6,053 held-out stays, 449 with events. All heads also validated on an external hospital (eICU) they never trained on.

GENERATIVE AI, KEPT HONEST

Granite 4.0 writes the handover on LinuxONE, and we check it before anyone reads it.



- **Rules build the facts** (SBAR: situation, background, assessment, recommendation) straight from the data, so they are always right.
- **Granite writes** a short spoken summary with llama.cpp built for zDNN, in a container with no published port.
- **Two checks:** every number must appear in the facts, and a predicted risk may never be written as a diagnosis.
- If a check fails, the summary is **held back** and the rule-based text is shown. The same pipeline writes a **ward shift summary** over the last n hours.

AI SECURED · INNOVATION WITHOUT EXPOSURE

Hospitals train one model together without sharing a single patient record.

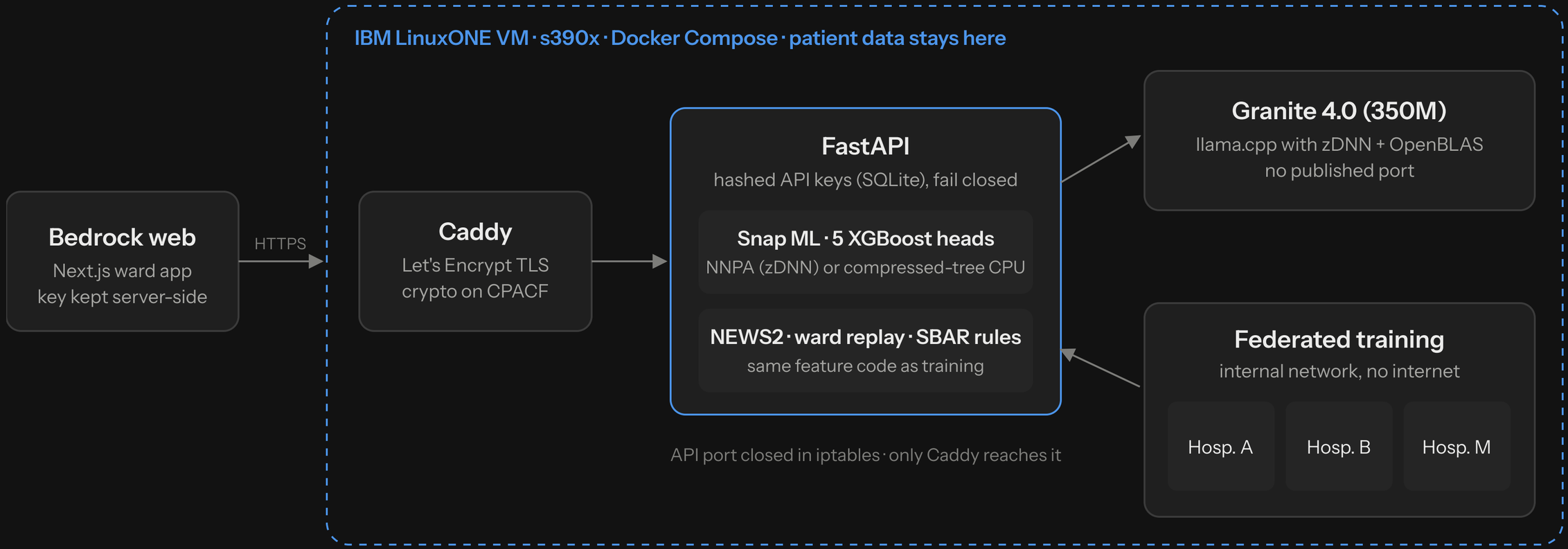
- Each hospital runs in its **own container on LinuxONE**, mounting only its own data, read-only, on an internal network with no internet.
- The model travels between hospitals (cyclic federated XGBoost). Each adds trees on its own patients and passes it on.
- A **privacy dial** adds Laplace noise to every new leaf (differential privacy) before the model leaves the building.
- The coordinator only ever sees models, held-out metrics and feature medians.

SEPSIS MODEL	HOSP. A	HOSP. B	MEAN AUROC
A's data alone	0.829	0.760	0.795
B's data alone	0.785	0.876	0.830
Federated	**0.817**	**0.867**	**0.842**
Federated, ε = 1	0.808	0.867	0.837
Federated, ε = 0.3	0.775	0.796	0.785
Pooled data (not allowed in reality)	0.827	0.871	0.848

A model from one hospital drops on another's patients. The federated one works for both and gets within 0.006 of pooling the data.

TECHNICAL DESIGN ON IBM LINUXONE (S390X)

Everything runs on one LinuxONE Community Cloud VM.



USING THE PLATFORM, AND MEASURING IT

We benchmarked the AI accelerator instead of assuming it helps.

SEPSIS HEAD, 3-CORE VM	1 ROW	WARD (12)	10K ROWS
Snap ML, NNPA (zDNN tensors)	0.089 ms	0.143 ms	30.7 ms
Snap ML, CPU (compressed trees)	**0.013 ms**	**0.043 ms**	**5.8 ms**
XGBoost 2.0.3, CPU	0.212 ms	0.208 ms	12.1 ms

All backends match the trained model within 5e-7. A whole ward is scored in well under a millisecond, so alarms are truly real time.

- **Snap ML** serves the trees; for our 100–575 depth-6 trees its CPU path beat NNPA, so we serve on CPU and keep NNPA one setting away.
- **zDNN on NNPA** runs Granite's llama.cpp for the handover text.
- **CPACF** handles the TLS crypto.
- **Containers** isolate each hospital, the model and the API, with no published ports except HTTPS. Granite runs byte-swapped for big-endian s390x.

0.043 ms
to score a whole 12-bed ward

1.7 M rows/s
throughput on 3 shared cores

~15 s
per Granite handover, generated on the VM

VALUE TO THE COMMUNITY

Who benefits, and what it takes to get there.

Patients

Earlier warning of sepsis, low blood pressure, kidney injury and breathing failure, when treatment works best.

Nurses and doctors

Fewer false alarms, a ranked ward, and a handover that is written for them and checked before they see it.

Hospitals

Better models from shared learning without moving patient data, on hardware built for regulated, always-on work.

Honest limits

- Retrospective research data, not a clinical trial. Decision support only: clinicians confirm.
- Per-tree ε adds up across trees, so the privacy dial shows the trade-off, not a certified guarantee.

Next steps

- Live HL7/FHIR vitals feed, SSO and per-user audit for a real trust.
- Prospective silent trial on one ward, then more hospitals in the federation.

BUILT DURING THE IBM Z DATATHON, 17–18 OCTOBER 2026

Bedrock: earlier warnings, fewer alarms, no data exposed.

Links

Code: **github.com/RayNCL2023/IBMZ-Datathon**
Live API: **148-100-78-203.sslip.io** (LinuxONE Community Cloud)

Data

PhysioNet/CinC Challenge 2019 · PhysioNet/CinC Challenge 2012 · eICU Collaborative Research Database demo (physionet.org)

Open source we built on

IBM Snap ML · XGBoost · scikit-learn · NumPy · pandas · FastAPI · Uvicorn · IBM Granite 4.0 · llama.cpp (zDNN build, IBM LinuxONE image) · gguf · Caddy · Docker · React · Next.js App Router on Vinext/Vite · Tailwind CSS · Remotion · Instrument Sans